

CONTROL STRUCTURES

```
mysql> CREATE TABLE student (  
  ->     id INT,  
  ->     name VARCHAR(50),  
  ->     mark INT  
  -> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> INSERT INTO student VALUES  
  -> (1, 'Ravi', 85),  
  -> (2, 'Priya', 65),  
  -> (3, 'Arun', 35);  
Query OK, 3 rows affected (0.01 sec)  
Records: 3  Duplicates: 0  Warnings: 0  
  
mysql> DELIMITER //  
mysql>  
mysql> CREATE PROCEDURE check_result(IN m INT)  
  -> BEGIN  
  ->     IF m >= 50 THEN  
  ->         SELECT 'PASS' AS Result;  
  ->     ELSE  
  ->         SELECT 'FAIL' AS Result;  
  ->     END IF;  
  -> END //  
Query OK, 0 rows affected (0.01 sec)  
  
mysql>  
mysql> DELIMITER ;  
mysql> CALL check_result(85);  
+-----+  
| Result |  
+-----+  
| PASS   |  
+-----+
```

```
mysql> DELIMITER //  
mysql>  
mysql> CREATE PROCEDURE grade(IN m INT)  
  -> BEGIN  
  ->     CASE  
  ->         WHEN m >= 90 THEN  
  ->             SELECT 'A Grade' AS Grade;  
  ->         WHEN m >= 75 THEN  
  ->             SELECT 'B Grade' AS Grade;  
  ->         WHEN m >= 50 THEN  
  ->             SELECT 'C Grade' AS Grade;  
  ->         ELSE  
  ->             SELECT 'Fail' AS Grade;  
  ->     END CASE;  
  -> END //  
Query OK, 0 rows affected (0.01 sec)  
  
mysql>  
mysql> DELIMITER ;  
mysql> CALL grade(82);  
+-----+  
| Grade |  
+-----+  
| B Grade |  
+-----+  
1 row in set (0.00 sec)
```

```
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE print_numbers()
-> BEGIN
-> DECLARE i INT DEFAULT 1;
-> WHILE i <= 5 DO
-> SELECT i;
-> SET i = i + 1;
-> END WHILE;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL print_numbers();
+----+
| i  |
+----+
| 1  |
+----+
1 row in set (0.00 sec)

+----+
| i  |
+----+
| 2  |
+----+
1 row in set (0.00 sec)

+----+
| i  |
+----+
| 3  |
+----+
1 row in set (0.00 sec)

+----+
| i  |
+----+
| 4  |
+----+
1 row in set (0.00 sec)

+----+
| i  |
+----+
| 5  |
+----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE repeat_numbers()
-> BEGIN
-> DECLARE i INT DEFAULT 1;
-> REPEAT
-> SELECT i;
-> SET i = i + 1;
-> UNTIL i > 5
-> END REPEAT;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE PROCEDURE repeat_numbers()
-> BEGIN
-> DECLARE i INT DEFAULT 1;
-> REPEAT
-> SELECT i;
-> SET i = i + 1;
-> UNTIL i > 5
-> END REPEAT;
-> END //
ERROR 1304 (42000): PROCEDURE repeat_numbers already exists
mysql> DELIMITER ;
mysql> CALL repeat_numbers();
+----+
| i  |
+----+
| 1  |
+----+
1 row in set (0.00 sec)

+----+
| i  |
+----+
| 2  |
+----+
1 row in set (0.00 sec)

+----+
| i  |
+----+
| 3  |
+----+
1 row in set (0.01 sec)

+----+
| i  |
+----+
| 4  |
+----+
1 row in set (0.01 sec)

+----+
| i  |
+----+
| 5  |
+----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```



Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //

mysql>

mysql> CREATE PROCEDURE loop_numbers()

-> BEGIN

-> DECLARE i INT DEFAULT 1;

->

-> number_loop: LOOP

-> SELECT i;

-> SET i = i + 1;

->

-> IF i > 5 THEN

-> LEAVE number_loop;

-> END IF;

-> END LOOP;

-> END //

Query OK, 0 rows affected (0.01 sec)

mysql>

mysql> DELIMITER ;

mysql> CALL loop_numbers();

i
1

1 row in set (0.00 sec)

i
2

1 row in set (0.00 sec)

i
3

1 row in set (0.01 sec)

i
4

1 row in set (0.01 sec)

i
5

1 row in set (0.01 sec)

Query OK, 0 rows affected (0.04 sec)

mysql> |